MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 4703: NETWORK ADMINSTRATION AND INFORMATION SECURITY

SUPPLIMENTARY EXAMINATION

DATE: 22nd SEPTEMBER, 2016 **TIME ALLOWED**: 2PM-5PM

INSTRUCTIONS:

i. Attempt **ALL** questions
ii. Write your answers clearly
iii. Marks for each question are indicated

QUESTION ONE: GENERAL NETWORKS

a) Define Network Security **[5 Marks]**
b) Explain perimeter defense as a network security measure. **[5 marks]**
c) Cryptography is not security. Discuss **[10 marks]**
d) Symmetric-Key and Asymmetric-Key Cryptography complement each other. Explain. **[5 marks]**
e) Describe **TWO** major security risks associated with wireless local area networks and measures that should be taken to reduce or eliminate them. **[10 Marks]**
f) Explain how email can be used to the recipients' disadvantage. **[5 Marks]**

g) You work in a multi-branched company. You have been given a task to assist management in trimming unnecessary ICT-related costs (Travel expenses, phone) as much as possible. As a Network Administrator with sound and updated knowledge, what would you do? **[10 Marks]**

h) Explain how Cryptography achieves the security goals. **[10 Marks]**

i) Suppose Alice wants to send "love u" to Bob using shift cipher using key=15. Calculate the cipher text that Bob will receive. **[5 Marks]**

QUESTION TWO: ADMINISTERING GLOBAL NETWORKS

a) Companies that have offices in several countries around the globe need to create a private corporate network that is able to connect these sites together and transport traffic of different types between them. An increasingly popular way of providing such a network is to use Multiprotocol Label Switching (MPLS) data services. Briefly explain how MPLS works and how it is able to support different traffic types. **[10 marks]**

b) An alternative to MPLS might be to consider using the Internet. What are the main disadvantages of the Internet that mean it would offer a worse solution than MPLS? **[5 marks]**

QUESTION THREE: ROUTING PROTOCOLS

a) Briefly describe the link-state protocol known as OSPF and explain how it copes with routing inside a large and complex autonomous system. **[10 marks]**

b) Remote access is often used by system administrators in large organisations to remotely control the computers on the organisation's local area network. Explain how malicious remote access administration tools (RATs) can be used. **[10 Marks]**

END OF QUESTION PAPER